

Name #1: _____

Name #2: _____

In-class Quiz

#06: Conditional Branching and Execution

Initialization

1. Create a constant string named **TESTDATA1** and give it the value "QUIZ06". (Don't forget to terminate your string).

2. Create a constant vector named **TESTDATA2** with the elements 5,7,1,3,2 (all elements are 2 byte).

3. Create a constant vector named **TESTDATA3** with the elements 9,2,4,8,3 (all elements are 4 bytes).

4. Create a 4-byte variable named **RESULT**.

Coding

5. Write the single instruction to ADD R2 to R3 and place the result in R1 only if the carry flag is set (=1). Update the flags in the instruction.

- Write the single instruction to SUB R2 from R3 and place the result in R1 only if the negative flag and the overflow flag are the same ($N=V$).
- Assume that register 0 (R0) has the value 0 and register 1 (R1) has the address of the constant string **TESTDATA1**. Write the code to find the location of letter "Z" in **TESTDATA1**. Place the result in R0.
- Give the final value (in 32-bit hexadecimal) of register 0 and 2 after the following code is executed. Assume that all condition flags are clear ($=0$) before beginning execution.

R0 = _____
R2 = _____

```

; Define constant
Input      DCD    4, 8
...
; Code
            MOV     R0, #0;
            LDR     R3, =Input
            LDR     R1, [R3]
            LDR     R2, [R3,#4]

lbegin
            CBZ     R2, lend

```

```
        ADD    R0, R1
        SUB    R2, #1
        B      lbegin
lend
```

9. Convert the following C code snippets to ARM assembly. Assume all variables are 32 bits. You do not have to define the variables, but you must load/store their values from/to RAM. You are NOT allowed to use any Branch instructions for this problem.

```
switch(VarX) {
    case 1:
        VarY = VarA;
        break;
    case 2:
        VarY = VarB;
        break;
    default:
        VarY = VarC;
        break;
}
```

10. Write the assembly code to count how many numbers in **TESTDATA2** are larger than the corresponding numbers in **TESTDATA3**. (ex. The 2nd number in TESTDATA2 is larger than the 2nd number in TESTDATA3). Place the result in the variable **RESULT**. (Be sure to note the size of the different data elements.) You MUST use a loop to go through all elements in the vector. (The constants and variable have already been defined and initialized on page 1.)